Northgate Distribution Warehouse - Structural Calculations

Document NDW-CALC-001, Revision B, January 2024. Project: Northgate.
SYNTHETIC TEST DATA - NOT FOR ENGINEERING USE

1 Scope

Structural design of a single-storey steel portal frame warehouse with an internal storage mezzanine and a two-storey office block.

2 Design standards

Eurocodes EN 1990, EN 1991 and EN 1993 with the UK National Annexes.

3 Design loads

Area	Imposed load	Notes
Roof (no access)	0.6 kN/m²	Category H
Mezzanine floor	7.5 kN/m²	Storage, category E1
Office floor	2.5 kN/m²	Category B
Roof snow	0.5 kN/m²	Characteristic value

3.1 Roof load limit

The roof is designed for an imposed load of 0.6 kN/m². It is not designed for plant or solar panels. Any additional roof load needs a new structural assessment.

4 Materials

Element	Specification
Portal frames	S355 hot-rolled structural steel
Purlins	S390GD cold-formed galvanised steel
Floor slab	C30/37 concrete with B500B reinforcement

4.1 Foundations

Pad foundations. Allowable bearing pressure: 150 kPa, from the geotechnical report NDW-GI-02.

5 Compliance

Building control approval status: PENDING. Submitted on 2 February 2024, reference BC-NDW-0042. The fire engineering review is still outstanding.